



FrameTracer

Multimodal AI Video Verification System for Digital Safety in Bangladesh

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Problem Statement & Context



Digital Expansion in Bangladesh

Rapid growth in smartphone adoption and social media usage (Facebook, YouTube, TikTok, WhatsApp, Telegram) has greatly expanded information reach across rural and urban populations.



Misinformation & Threat Risks

Manipulated and AI-generated short videos spread fast, triggering public panic, communal tension, political instability, financial scams, and fake disaster alerts before manual verification can occur.

Proposed Solution & Value

Explainable Risk Assessment

- FrameTracer provides a web-based decision-support platform where users upload suspicious short videos for immediate automated evaluation.
- Rather than a binary "fake or real" output, it delivers an **explainable multi-factor risk score** categorized into Low, Medium, or High Risk, aiding journalists and general citizens alike.



High Risk



Low Risk

System Technical Workflow



1. Video Upload

User submits short video ($\leq 30s$) via Next.js frontend to Supabase Storage & PostgreSQL metadata logging.



2. Preprocessing

FFmpeg background worker processes video file to extract keyframes and separate audio streams.



3. Multimodal Analysis

Gemini Vision API (frames), Google Video Intelligence (shots/labels), and Whisper API (audio transcripts).



4. Synthesis Report

LLM engine correlates visual & transcript indicators to generate human-readable explanations & risk scores.

Multimodal AI Analysis

Visual Artifact Detection

Analyzes frame boundaries for unnatural face swaps, inconsistent lighting/shadows, artificial skin textures, distorted limbs/hands/eyes, and misleading text overlays.

Audio & Context Analysis

Transcribes speech via Whisper API to detect robotic inflection, synthetic speech patterns, unnatural pauses, and compares context against known regional scam templates.

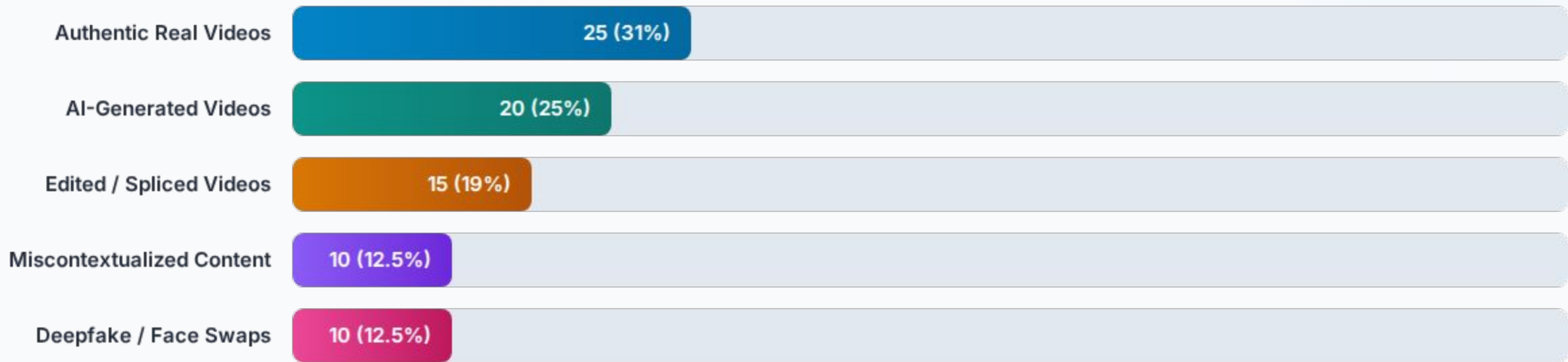
Pilot Implementation Scope

- ✓ **Secure User Portal:** Authentication & video history dashboard powered by Supabase.
- ✓ **Short Video Focus:** Supports videos up to 30 seconds for optimal pilot processing speed (< 5 mins).
- ✓ **Automated Processing Pipeline:** Pre-integrated FFmpeg frame cutter and multi-API analysis engine.
- ✓ **Explainable Output:** Generates flagged keyframe previews, audio transcripts, and plain-language risk breakdowns.
- ✓ **Evaluated Pilot Benchmark:** Test against a curated 80-video dataset representing local context.

Recommended Tech Stack

System Layer	Recommended Technology	Function / Role
Frontend Framework	React, Next.js (App Router), Tailwind CSS	User Dashboard & Upload Interface
Backend & Auth	Supabase Auth & PostgreSQL	User Identity & Metadata Management
Video Preprocessing	FFmpeg / Python Worker	Frame Extraction & Audio Channel Separation
Vision & Audio AI	Gemini Vision, Google Video Intel, Whisper	Multimodal Deepfake & Content Analysis
Report Generation	Gemini Pro / GPT-4o	Contextual Synthesis & Risk Scoring

Dataset & Evaluation Metrics



Curated 80-video pilot evaluation benchmark targeting 70% – 85% detection accuracy with processing times under 5 minutes per video.

Budget & Resource Allocation

163,000

Total Pilot Budget (BDT)

Covers 2-Month Pilot Phase & API Credits

Category	Cost (BDT)
AI APIs (Gemini, Google Video, Whisper)	20,500 BDT
Cloud & Database (Supabase, Vercel, R2)	20,500 BDT
Development & AI Coding Assistance	60,000 BDT
Dataset Collection & Labeling	50,000 BDT
Contingency Buffer (10%)	12,000 BDT

Social Impact & SDG Alignment



SDG 16 (Peace & Justice): Combats malicious digital media manipulation, protecting public trust and societal harmony.



SDG 9 (Innovation): Fosters localized AI digital safety tools tailored specifically for Bangla language context.



Public Emergency Safety: Mitigates panic caused by false disaster alerts, scams, or doctored political videos.

Business & Sustainability



B2B / B2G API Access

Monetized API subscriptions for newsrooms, NGOs, fact-checking organizations, and social networks.



Enterprise Dashboard

Custom monitoring for high-profile public events, election integrity, and disaster management teams.

Pricing & Subscription Plans

Free Tier

0 BDT /month

- ✓ 3 checks / month
- ✓ Web platform access
- ✓ Standard analysis speed

Get Started

Basic Plan

POPULAR

500 BDT /month

- ✓ 20 checks / month
- ✓ High-speed processing
- ✓ Email support access

Choose Basic

Pro Plan

2,500 BDT /month

- ★ 150 checks / month
- ★ Priority verification queue
- ★ 24/7 Premium live support

Upgrade to Pro

Project Conclusion & Future Outlook



Strengthening Digital Trust

By automating the verification process, FrameTracer drastically reduces response times for detecting altered media, providing crucial protection against rapidly spreading online misinformation in Bangladesh.



Explainable AI & Accessibility

Rather than a opaque yes/no decision, transparent risk metrics enable local fact-checkers, journalists, and citizens to understand exactly why and how a video might be manipulated.



Future Enhancements

Plans include expanding deepfake classification support, optimizing real-time threat detection, and integrating localized language processing models tailored to regional dialects.

[Complete Overview](#)



Thank You